



Lesson 2.2 — Vertex Form and Transformations

Big idea: Vertex form $y = a(x - h)^2 + k$ tells you the vertex and the transformation from $y = x^2$ at a glance — (h, k) .

Quick review

Vertex form: $y = a(x - h)^2 + k$. Vertex is (h, k) .

Transformation read: shift right h , up k ; vertical stretch by a (flip if $a < 0$).

From standard → vertex: complete the square OR use $h = -b/(2a)$ and $k = f(h)$.

From vertex → standard: just expand.

Worked example

Worked example. Convert $y = x^2 - 6x + 11$ to vertex form.

$h = 6/2 = 3$, $k = (3)^2 - 18 + 11 = 2$. So $y = (x - 3)^2 + 2$. (Or complete the square: $x^2 - 6x = (x - 3)^2 - 9$; add 11 $\rightarrow (x - 3)^2 + 2$.)

Warm-ups

1. State the vertex of $y = (x - 4)^2 + 1$.
2. State the vertex of $y = -2(x + 5)^2 - 3$.
3. Write a parabola in vertex form with vertex $(1, -7)$ and $a = 1$.

Practice

1. Convert $y = x^2 + 8x + 10$ to vertex form.
2. Convert $y = 2x^2 - 12x + 7$ to vertex form.
3. Describe the transformations from $y = x^2$ to $y = -3(x + 2)^2 + 4$.

Challenge

1. A parabola passes through $(0, 8)$ and has vertex $(3, -1)$. Write its equation in vertex form.
2. Convert $y = -x^2 + 4x - 7$ to vertex form, then state the maximum value of the function.



Answer key — Lesson 2.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. (4, 1).
2. (-5, -3).
3. $y = (x - 1)^2 - 7$.

Practice

1. $x^2 + 8x = (x + 4)^2 - 16 \Rightarrow y = (x + 4)^2 - 6$.
2. $h = 12/4 = 3$, $k = 2(9) - 36 + 7 = -11$. $y = 2(x - 3)^2 - 11$.
3. Shift left 2, up 4; vertical stretch by 3; reflect across x-axis.

Challenge

1. $y = a(x - 3)^2 - 1$. Plug (0, 8): $8 = 9a - 1 \Rightarrow a = 1$. $y = (x - 3)^2 - 1$.
2. $y = -(x^2 - 4x) - 7 = -[(x - 2)^2 - 4] - 7 = -(x - 2)^2 - 3$. Maximum value = -3 (at $x = 2$).